Neuron mechanisms IV: amplifiers pass parities through the linear window of relu(tanh)

wd3_k0_N2048, step 500k — post-nonlinearity units, uniform evaluation data

The puzzle and the measurement

Amplifier neurons (blocks 2–3) receive an already-assembled parity: at least 90% of their pre-activation variance lies on their own χ_{d8} (276 active neurons) or χ_{d4} (22), and they re-emit that same mode at high purity. Since the parity is a ± 1 signal, one would expect $\phi = \mathrm{relu}(\tanh)$ to cut the negative half and distort the feature. It does not, because the input is not centered. Writing the pre-activation on the own parity χ as $z = c + \gamma\chi$ and measuring both constants per neuron gives Table 1 and Fig. 1: **the DC offset exceeds the mode amplitude**, $c > |\gamma|$, for 297 of 298 amplifiers (median $c/|\gamma| \approx 1.4$). Both conditional levels $c \pm |\gamma|$ are therefore positive — z never reaches the relu corner. Moreover the levels are small: the 95th-percentile upper level is 0.11 ($d8$) and 0.22 ($d4$), where tanh deviates from the identity by $z^2/3 \leq 0.4\%/1.6\%$. So on its operating window the neuron is a linear wire, $a \approx c + \gamma\chi$.

	n	med. c	med. $\lvert\gamma\rvert$	$c > \lvert\gamma\rvert$	z 95%	$\frac{\Delta a}{\Delta z}$	$\frac{\lvert\langle a\chi\rangle\rvert}{\lvert\gamma\rvert}$	vs. clipped $\frac{\tanh\gamma}{2}$
$d8$ amplifiers	276	0.040	0.028	275/276	0.106	0.996	0.995	1.99×
$d4$ amplifiers	22	0.086	0.061	22/22	0.218	0.991	0.992	1.99×

Table 1: Per-neuron fits of $z = c + \gamma\chi$ and linearity checks. $\Delta a/\Delta z$: output swing over input swing between the two parity classes; $|\langle a\chi\rangle|/|\gamma|$: output over input Walsh coefficient. Both are ≈ 1, i.e. twice what half-wave rectification of a centered input would leave.

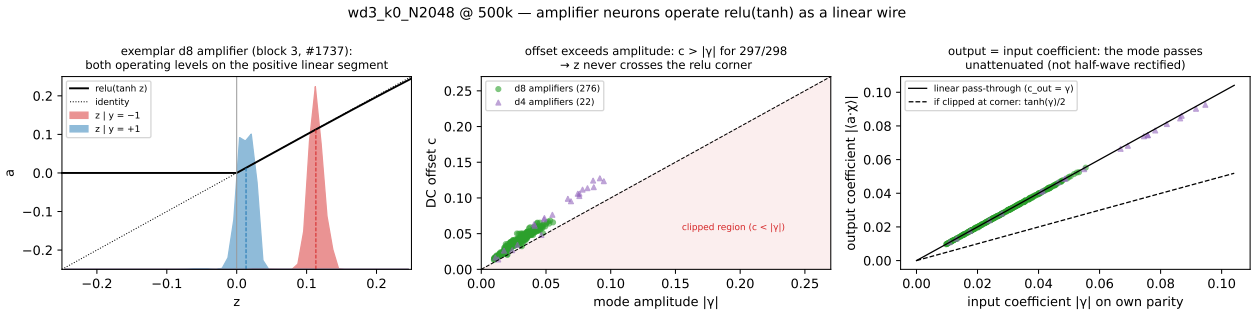


Figure 1: Left: exemplar $d8$ amplifier — the two conditional distributions of z (for $\chi = \pm 1$) both sit on the positive segment of relu(tanh), where the curve is indistinguishable from the identity. Middle: population scatter of offset c vs. amplitude $|\gamma|$; every neuron but one lies above the $c = |\gamma|$ line, outside the clipped region. Right: output vs. input coefficient — the points follow the identity (linear pass-through), not the half-wave prediction $\tanh(\gamma)/2$ (dashed).

Why the offset is there, and what "amplification" means

The offset is not a learned bias (the block linear layers have none). It is inherited from the previous rectification: upstream carriers of a parity are one-sided units with activations $\approx t(1 + \chi)/2$, i.e. each delivers the mode with a DC component of the same size attached. Any weighted sum of such features arrives with $c \sim |\gamma|$ automatically — the DC produced by one layer's relu is exactly what shields the feature from the next layer's relu.

The gain is therefore not pointwise ($|\langle a\chi\rangle| = |\gamma|$, nothing is amplified inside the neuron). The amplification is collective: the V-row coherently sums many weak copies of χ spread across the 2048 residual channels into a single clean γ (reading the stream is non-destructive), and the neuron writes a fresh copy back at purity 0.94/0.92 (median share of Var a on the own mode) — versus 0.044 for the raw-bit gates that created $d8$. Hundreds of such units re-emitting in parallel grow the total mode power of the stream from block to block. Both polarities are present (148+/128− for $d8$, 8+/14− for $d4$), so downstream sums can recover the full ± 1 swing and cancel the accumulated DC. This completes the contrast with the creator mechanisms: gates and corner units sit *on* the relu corner and use it to compute new modes, while amplifiers are biased *past* the corner so as to transport existing modes unchanged.

Artifacts: `gdrive:parity/analysis_output/neuron_mechanisms/wd3_k0_N2048/step_00500000/` (`amplifier_stats.csv`, this figure and report; companion $d2/d4/d8/d16$ reports in the same folder).